Using Tabular Editor

For faster Power BI development

Kimura
data intelligence

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Disclaimer

- I am not affiliated with Tabular Editor

- I just think the tool is nice for Power BI developers that want to work smarter, not harder

Slides and code samples

- The slides and code samples for this session will be posted to my public Github repo:
- https://github.com/basland/presentations/

- Also check my blog for content like this:
- https://thatfabricguy.com

Agenda

- Problem with manual development in Power BI semantic models
- Introduction of Tabular Editor
- Naming conventions

- Creating relationships
- Bulk measure creation

- Best practice analyzer
- One secret goodie ☺

About Tabular Editor

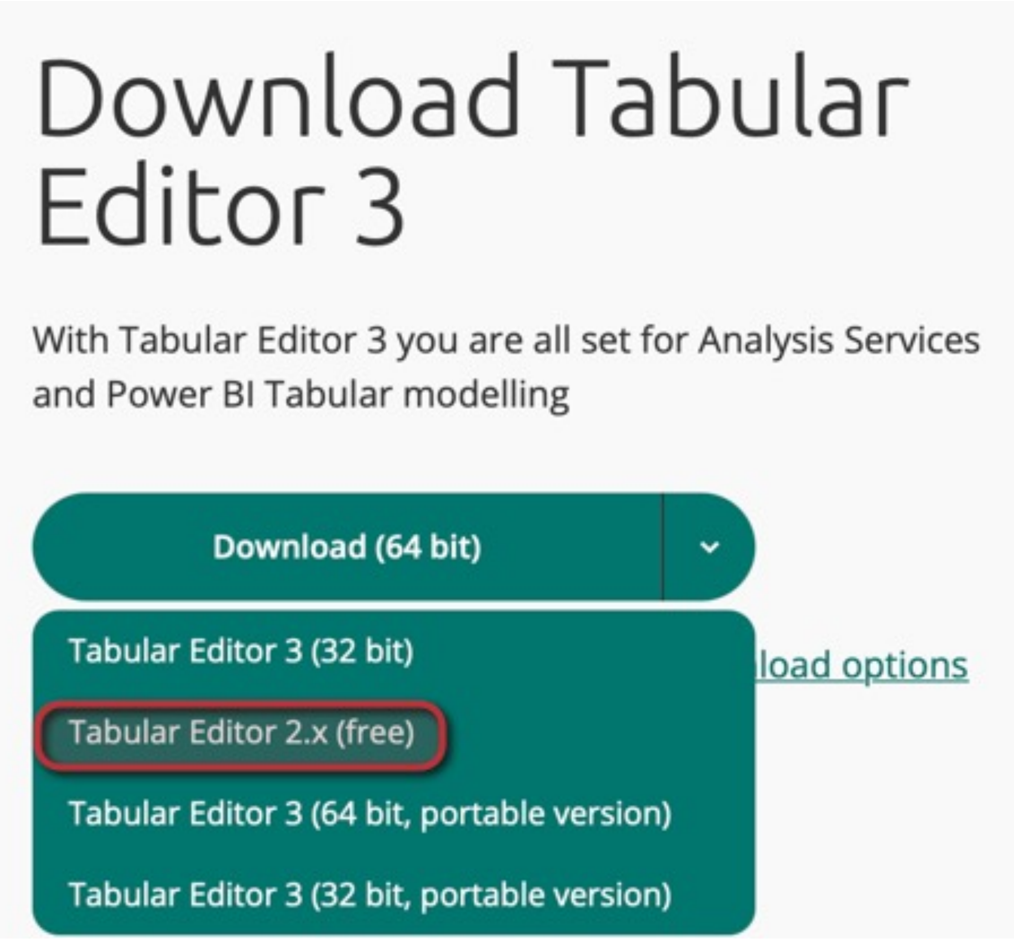

THIS
MIGHT
TAKE
A WHILE

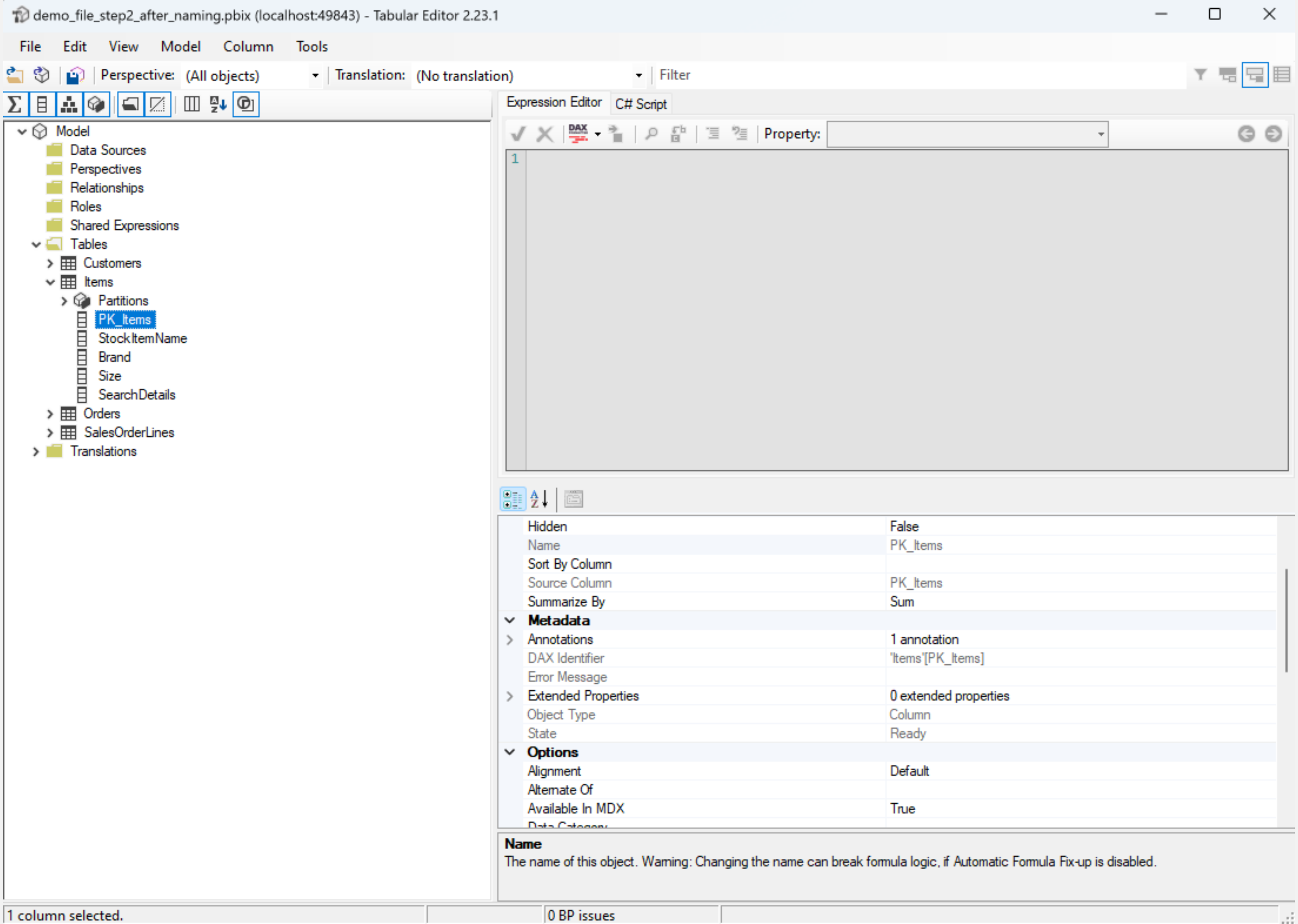
Tabular Editor

- Started as open source software
- Currently a for profit (and rightfully so!)
- Helps edit Power BI and Analysis Services models
- Works alongside Power BI Desktop and Service
- Both free version (v2) and paid (v3) available
- Today's demo will use v2

How to download?

tabulareditor.com/downloads




demo_file_step2_after_naming.pbix (localhost:49843) - Tabular Editor 2.23.1
File
Edit
View
Model
Column
Tools
Perspective: (All objects)
Translation: (No translation)
Filter
Model
Data Sources
Perspectives
Relationships
Roles
Shared Expressions
Tables
Customers
Items
Partitions
PK_Items
StockItemName
Brand
Size
SearchDetails
Orders
SalesOrderLines
Translations
Expression Editor
C# Script
Property:
Hidden
False
Name
PK_Items
Sort By Column
Source Column
PK_Items
Summarize By
Sum
Metadata
Annotations
1 annotation
DAX Identifier
'Items'[PK_Items]
Error Message
Extended Properties
0 extended properties
Object Type
Column
State
Ready
Options
Alignment
Default
Alternate Of
Available In MDX
True
Name
The name of this object. Warning: Changing the name can break formula logic, if Automatic Formula Fix-up is disabled.
1 column selected.
0 BP issues

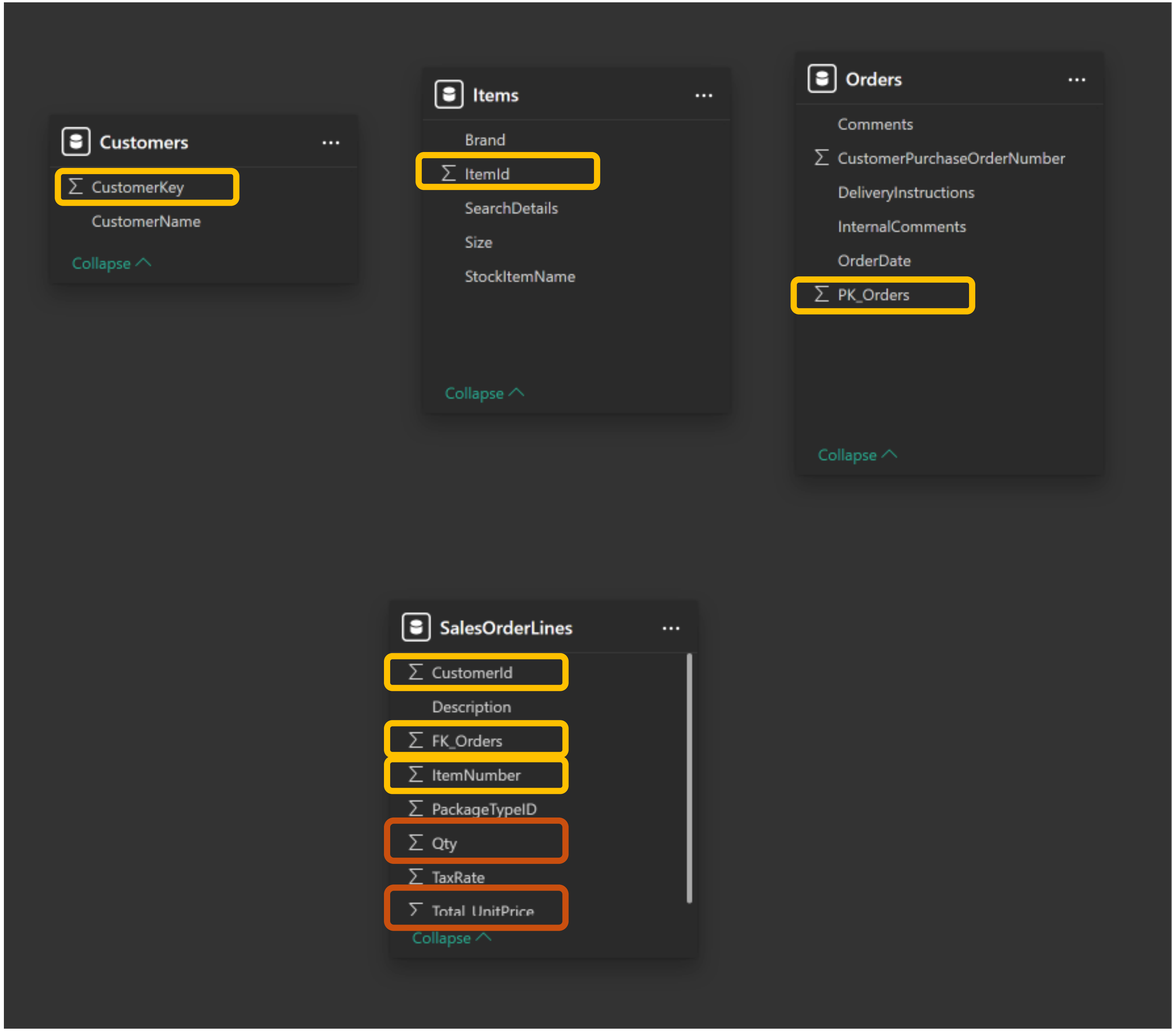
Theoretical tips ‘n tricks

Naming conventions

- Generally **agreed scheme** for naming things
- Makes collaborating with colleagues easier
- Also makes collaborating with automation tools easier!
- Use prefixes or suffixes for column names
- At the minimum: relationship columns and measure columns

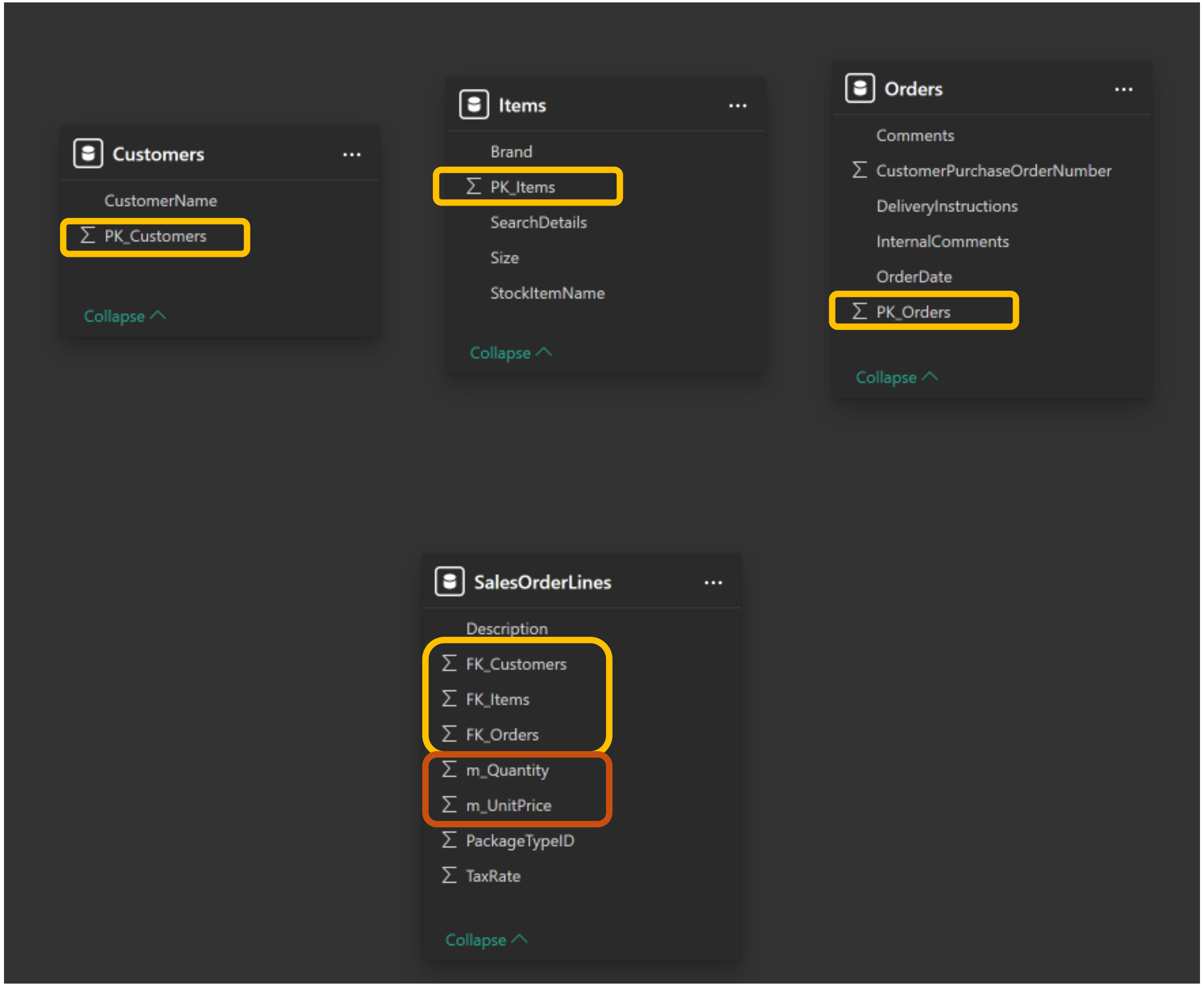
Bad example

- Foreign keys are all over the place!
- No standard measure naming



Good example

- FK_ and PK_ for relationship key columns
- M_ for measure columns



Automations with TE2

- Programmatically create objects
- Use naming conventions to bulk create objects
- Easily cleanup your model afterwards
- Write C# scripts to automate tasks
- I’m not that good with C#, but you know who is?
- Bloggers.
- Youtubers.
- ChatGPT!

Automation examples

Creating relationships

Expression Editor C# Script

Samples 100 %

```csharp
// Example for Power BI Gebruikersdagen 2025
// Automating Power BI development with Tabular Editor
// Created by Bas Land, Kimura Data Intelligence B.V. and ThatFabricGuy.com


// Loop through all tables to find PK_ columns
foreach (var pkTable in Model.Tables)
{
    foreach (var pkColumn in pkTable.Columns)
    {
        if (pkColumn.Name.StartsWith("PK_"))
        {
            // Extract the key name (without the "PK_" prefix)
            var keyName = pkColumn.Name.Substring(3);

            // Search for corresponding FK_ column in other tables
            foreach (var fkTable in Model.Tables)
            {
                if (fkTable == pkTable) continue; // Skip itself (dont create relationship with self

                foreach (var fkColumn in fkTable.Columns)
                {
                    if (fkColumn.Name == "FK_" + keyName)
                    {
                        // Check if relationship does not already exist
                        bool relationshipExists = Model.Relationships
                            .Any(r => r.ToColumn == pkColumn && r.FromColumn == fkColumn);

                        if (!relationshipExists)
                        {
                            // Create the relationship
                            var rel = Model.AddRelationship();
                            rel.FromColumn = fkColumn;  // Foreign key is always on the "From" side (fact table)
                            rel.ToColumn = pkColumn;    // Primary key is on the "To" side (dimension table)
                            rel.IsActive = true;
                            rel.CrossFilteringBehavior = CrossFilteringBehavior.OneDirection; // Single-direction filtering

                            // Hide the key columns, we don't want to use them in the report
                            pkColumn.IsHidden = true;
                            fkColumn.IsHidden = true;
                        }
                    }
                }
            }
        }
    }
}
```

Automation examples

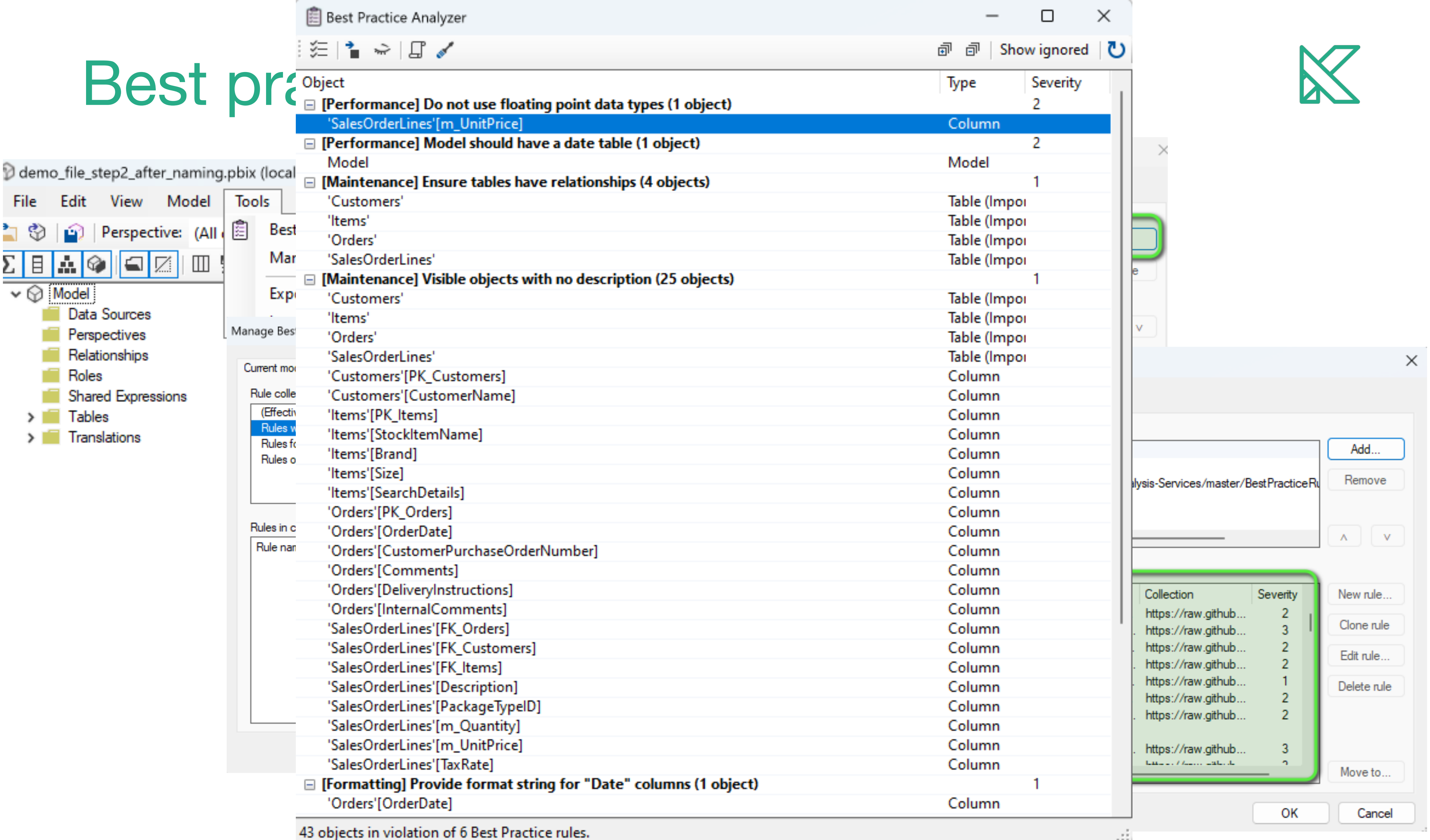
Creating measures

```
// Example for Power BI Gebruikersdagen 2025
// Automating Power BI development with Tabular Editor
// Created by Bas Land, Kimura Data Intelligence B.V. and ThatFabricGuy.com

// Define the table name
var measureTableName = "Calculations";

// Check if the table already exists
var measureTable = Model.Tables.FirstOrDefault(t => t.Name == measureTableName);

if (measureTable == null)
{
    // Create a calculated table using a simple DAX expression with a placeholder column
    string daxExpression = "DATATABLE(\"DummyColumn\", STRING, {{\"Placeholder\"}})";

    measureTable = Model.AddCalculatedTable(measureTableName, daxExpression);

    // Ensure DummyColumn is hidden
    var dummyColumn = measureTable.Columns.FirstOrDefault(c => c.Name == "DummyColumn");
    if (dummyColumn != null)
    {
        dummyColumn.IsHidden = true;
    }
}

// Loop through all tables to find m_ columns
foreach (var table in Model.Tables)
{
    foreach (var column in table.Columns.ToList()) // Convert to list to avoid modification issues
    {
        if (column.Name.StartsWith("m_"))
        {
            // Extract measure name (remove "m_" prefix)
            var measureName = column.Name.Substring(2);
            var daxExpression = "SUM ( " + table.Name + "[" + column.Name + "] )";

            // Check if the measure already exists
            var existingMeasure = measureTable.Measures.FirstOrDefault(m => m.Name == measureName);

            if (existingMeasure != null)
            {
                existingMeasure.Expression = daxExpression;
            }
            else
            {
                // Create new measure
                var newMeasure = measureTable.AddMeasure(measureName, daxExpression);
                newMeasure.DisplayFolder = table.Name;
            }

            // Hide the original column
            column.IsHidden = true;
        }
    }
}
```

Best practise analyzer

- Tool to help you follow best practices
- Fully automated
- Rule based
- Default rulesets available!
  - > https://powerbi.microsoft.com/nl-nl/blog/best-practice-rules-to-improve-your-models-performance/
  - Ruleset in a file: https://raw.githubusercontent.com/microsoft/Analysis-Services/master/BestPracticeRules/BPARules.json

Best pra



Best Practice Analyzer
Show ignored
Object
Type
Severity
[Performance] Do not use floating point data types (1 object)
2
'SalesOrderLines'[m_UnitPrice]
Column
[Performance] Model should have a date table (1 object)
2
Model
Model
[Maintenance] Ensure tables have relationships (4 objects)
1
'Customers'
'Items'
'Orders'
'SalesOrderLines'
Table (Impoı
[Maintenance] Visible objects with no description (25 objects)
1
'Customers'
'Items'
'Orders'
'SalesOrderLines'
'Customers'[PK_Customers]
'Customers'[CustomerName]
'Items'[PK_Items]
'Items'[StockItemName]
'Items'[Brand]
'Items'[Size]
'Items'[SearchDetails]
'Orders'[PK_Orders]
'Orders'[OrderDate]
'Orders'[CustomerPurchaseOrderNumber]
'Orders'[Comments]
'Orders'[DeliveryInstructions]
'Orders'[InternalComments]
'SalesOrderLines'[FK_Orders]
'SalesOrderLines'[FK_Customers]
'SalesOrderLines'[FK_Items]
'SalesOrderLines'[Description]
'SalesOrderLines'[PackageTypeID]
'SalesOrderLines'[m_Quantity]
'SalesOrderLines'[m_UnitPrice]
'SalesOrderLines'[TaxRate]
Column
[Formatting] Provide format string for "Date" columns (1 object)
1
'Orders'[OrderDate]
Column
43 objects in violation of 6 Best Practice rules.
demo_file_step2_after_naming.pbix (local
File
Edit
View
Model
Tools
Perspective: (All
Model
Data Sources
Perspectives
Relationships
Roles
Shared Expressions
Tables
Translations
Manage Best
Add...
Remove
New rule...
Clone rule
Edit rule...
Delete rule
Move to...
Collection
Severity
https://raw.github...
OK
Cancel

Secret topic

- I won’t tell you…
- Keep watching for the end of the demo!


Demo time

Conclusion

- Using Tabular Editor (even the free version) you can automate a lot
- Naming conventions are necessary to help automate
- Even AI can be utilised to quickly generate all kinds of content

Bas Land

- BI consultant since 2013
- Co-founder of Kimura Data Intelligence
  - Kimura Data Framework for Fabric
  - DataChimp for bookkeeping firms
  - Analytical products for several software partners
  - Consultancy (Fabric & Power BI)

- Married to Anouk, we have a daxhund (☺) called Chester, and a baby on the way

- Sports: purple belt Brazilian jiu-jitsu, weight lifting, running

Contact

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